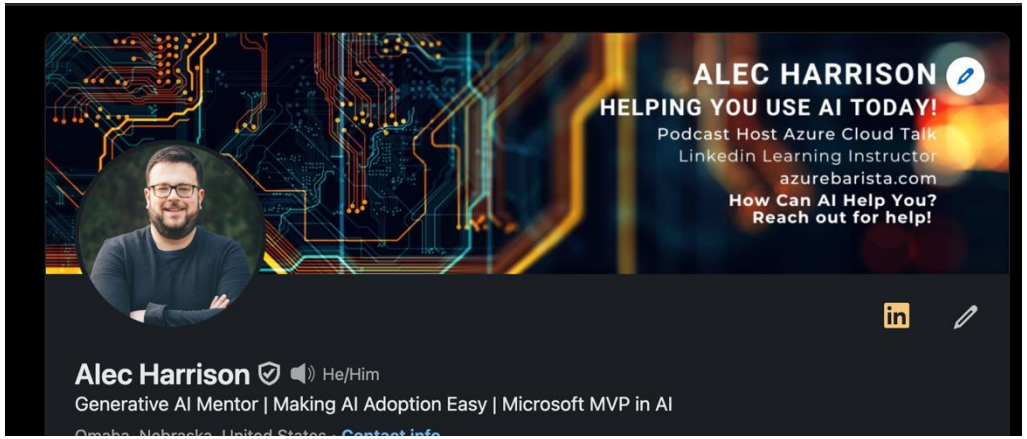
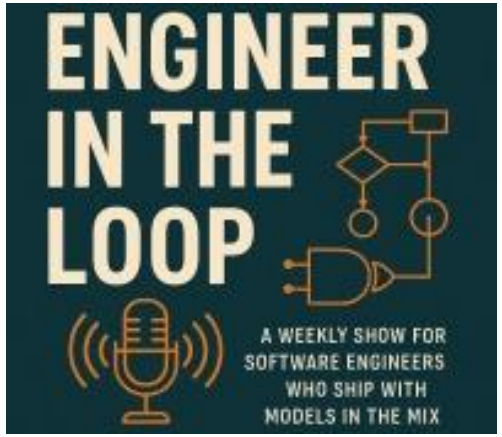
AI Foundry Local

BUILDING, RUNNING, AND GOVERNING AI LOCALLY

About Me

- Podcast Host
- Blogger (EITL.AI)
- Microsoft MVP in AI
- Microsoft Certified Trainer
- Gen AI Staff Engineer
- New Dad





LEAN TECHNIQUES

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MISSION

Helping companies make technology a competitive advantage.

AREAS OF FOCUS

Software • Cloud • AI • Data & Analytics
Product Management & Design • Tech Leadership

WE'RE HIRING!

Based in Midwest with a distributed workforce.
Currently hiring for a variety of positions!

Agenda

- Why AI Foundry Local
- What AI Foundry Local Is
- Architecture Overview
- Core Capabilities
- Use Cases
- Demo / Walkthrough
- Operational & Governance Considerations
- Getting Started

Why Run AI Locally?

- Data residency and compliance requirements
- Latency-sensitive workloads
- Cost control for high-volume inference
- Offline or edge scenarios
- Security and IP protection

What Is AI Foundry Local?

A local deployment of AI Foundry capabilities

Run models, tools, and workflows on your own infrastructure

Consistent experience with cloud-based AI Foundry

Designed for enterprise and regulated environments

Core Components

MODEL HOSTING AND INFERENCE RUNTIME

PROMPT AND WORKFLOW ORCHESTRATION

EVALUATION AND MONITORING

SECURITY AND ACCESS CONTROLS

INTEGRATION WITH EXISTING APPS AND DATA

High-Level Architecture

- Local compute (VMs, Kubernetes, or on-prem servers)
- Model registry and artifact storage
- API layer for apps and services
- Observability and logging
- Optional cloud connectivity

Model Support

- Open-source models (e.g., LLMs, embeddings)
- Fine-tuned or custom models
- Versioning and lifecycle management
- Bring-your-own-model flexibility

Prompt & Workflow Management

Centralized prompt templates

Reusable chains and tools

Environment-specific configurations

Version control for prompts

Evaluation & Quality

Offline and online evaluation

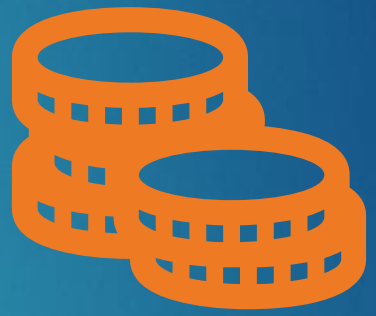
Regression testing for prompts and models

Human-in-the-loop review

Metrics for accuracy and relevance

Monitoring & Observability

Inference latency and throughput



Token usage and cost estimation ($0.00)

Security & Governance

Local data control

Role-based access control

Audit logging

Common Use Cases

- Internal copilots and assistants
- Document processing and summarization
- Code analysis and generation
- Edge or disconnected environments
- Regulated industry workloads

Example Workflow

User request hits internal app

App calls AI Foundry Local API

Model processes data locally

Response returned

Demo

Deploy	Deploy a local model
Create	Create a prompt
Run	Run an inference
Review	Review logs and metrics

Operational Considerations

- Hardware sizing and GPU selection
- Scaling strategies
- Upgrade and patching process
- Backup and recovery

Local vs Cloud AI Foundry

- Local: control, compliance, low latency
- Cloud: elasticity, managed services
- Hybrid approaches are common
- Choose per workload

Getting Started

- Identify target use case
- Select hardware and runtime
- Deploy AI Foundry Local
- Pilot with a small team
- Iterate and scale

Best Practices

- Start with evaluation early
- Treat prompts as code
- Monitor usage continuously
- Plan for governance from day one

Common Questions

- https://learn.microsoft.com/en-us/azure/ai-foundry/foundry-local/what-is-foundry-local?view=foundry-classic&viewFallbackFrom=foundry

Roadmap & Future

- Expanded model support
- Improved tooling and automation (prompty)
- Deeper hybrid integrations
- Community and ecosystem growth

Q&A

QUESTIONS

DISCUSSION

SHARE EXPERIENCES